# Empirical Validation of a Mathematical Framework of Housing

## Evidence from the United Kingdom

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

**Abstract**

This paper presents an empirical framework for validating a multidimensional mathematical model of housing using publicly available United Kingdom data. Housing is represented as a latent state-space consisting of structural, financial, ownership, legal, and occupancy components. The paper combines established techniques from housing economics, panel econometrics, state-space modelling, stock-flow consistent macroeconomics, and machine learning to construct a unified empirical representation of the UK housing market.

The paper ends with "The End"

# 1 Introduction

Housing occupies a central role within the United Kingdom economy through household wealth accumulation, mortgage finance, urban development, and financial stability. The UK housing market exhibits substantial regional heterogeneity, persistent supply constraints, and close interaction with monetary policy.

Building upon the companion theoretical framework, this paper demonstrates how the proposed multidimensional housing state-space can be estimated using publicly available UK datasets.

# 2 Empirical State-Space

For region $r$ and time $t$, define

$$X_{r,t} = (S_{r,t}, F_{r,t}, O_{r,t}, L_{r,t}, Q_{r,t}),$$

where

$S$ structural housing conditions,

$F$ financial conditions,

$O$ ownership composition,

$L$ legal and repossession conditions,

$Q$ occupancy conditions.

Observed variables satisfy

$$Y_{r,t} = HX_{r,t} + \varepsilon_{r,t},$$

while the latent state evolves according to

$$X_{r,t+1} = AX_{r,t} + BU_t + \eta_{r,t},$$

where $U_t$ contains macroeconomic variables including Bank of England policy rates, inflation, unemployment, and GDP growth.

## 3 United Kingdom Data

The empirical analysis may be implemented entirely from publicly available datasets.

| Variable | Representative Data Source |
|---|---|
| House prices | HM Land Registry UK House Price Index |
| Mortgage approvals | Bank of England |
| Mortgage interest rates | Bank of England |
| Building permits | Ministry of Housing, Communities and Local Government |
| Housing completions | Department for Levelling Up, Housing and Communities |
| Household income | Office for National Statistics |
| Unemployment | Office for National Statistics |
| Population | Office for National Statistics |
| Vacancy statistics | Office for National Statistics |
| Repossessions | UK Finance |

## 4 Variable Construction

| Latent Variable | Observable Proxy |
|---|---|
| Structural state | Housing completions, building permits |
| Financial state | Mortgage approvals, outstanding lending, mortgage rates |
| Ownership state | Homeownership rate, buy-to-let share |
| Legal state | Mortgage repossessions, insolvency measures |
| Occupancy state | Vacancy rates, rental share |

## 5 Econometric Framework

A benchmark panel regression is

$$\Delta P_{r,t} = \alpha_r + \lambda_t + \beta' Z_{r,t} + \varepsilon_{r,t},$$

where

$$Z_{r,t} = (\text{Income}, \text{Mortgage Rate}, \text{Housing Supply}, \text{Unemployment}, \text{Mortgage Lending}).$$

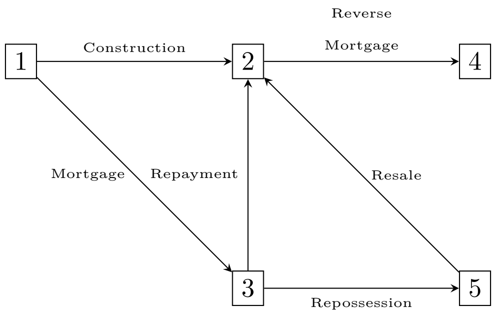
Mortgage distress may be modelled using

$$Pr(D_{r,t} = 1) = \Lambda(\beta' Z_{r,t}),$$

where $\Lambda(\cdot)$ denotes the logistic cumulative distribution function.

## 6 Housing Lifecycle

The theoretical lifecycle remains unchanged.



Empirical transition probabilities are estimated by

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_j n_{ij}},$$

where $n_{ij}$ denotes observed transitions.

## 7 Validation

The empirical study evaluates four hypotheses.

1. The multidimensional housing state-space is statistically identifiable.
2. Regional housing dynamics exhibit persistent transition structures.
3. Distinguishing financial, legal, ownership, and occupancy states improves predictive accuracy.
4. The proposed framework outperforms conventional house-price regressions.

Forecast performance may be evaluated using RMSE, MAE, log-likelihood, and out-of-sample prediction.

**Proposition 1.** *Conventional UK housing regressions are nested within the proposed multidimensional framework.*

*Proof.* If only observed house prices are retained and the remaining latent dimensions are omitted, the measurement equation reduces to a conventional reduced-form regression model. Therefore the multidimensional framework strictly generalizes standard empirical housing specifications. □

## 8 Discussion

The proposed framework is sufficiently general to incorporate classical panel-data methods, Bayesian estimation, hidden Markov models, Kalman filtering, graph-based models, and modern machine-learning algorithms. The latent-state representation also permits regional heterogeneity across England, Scotland, Wales, and Northern Ireland without changing the mathematical structure.

## 9 Conclusion

The paper presents a unified empirical architecture for analysing the United Kingdom housing market using publicly available data. The multidimensional state-space links housing supply, mortgage finance, ownership, occupancy, and legal status within a common mathematical framework. The modular formulation facilitates future extensions to regional housing policy, macroprudential regulation, affordability analysis, and AI-assisted housing forecasting.

## References

[1] Arrow, K. J., and Debreu, G. Existence of an Equilibrium for a Competitive Economy. *Econometrica*, 1954.

[2] Muth, R. F. *Cities and Housing*. University of Chicago Press, 1969.

[3] Rosen, S. Hedonic Prices and Implicit Markets. *Journal of Political Economy*, 1974.

[4] DiPasquale, D., and Wheaton, W. C. Housing Market Dynamics and the Future of Housing Prices. *Journal of Urban Economics*, 1994.

[5] Case, K. E., and Shiller, R. J. The Efficiency of the Market for Single-Family Homes. *American Economic Review*, 1989.

[6] Kiyotaki, N., and Moore, J. Credit Cycles. *Journal of Political Economy*, 1997.

[7] Godley, W., and Lavoie, M. *Monetary Economics*. Palgrave Macmillan, 2007.

[8] Puterman, M. *Markov Decision Processes*. Wiley, 1994.

[9] Harvey, A. C. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, 1989.

[10] Bertsekas, D. *Dynamic Programming and Optimal Control*. Athena Scientific.

[11] Russell, S., and Norvig, P. *Artificial Intelligence: A Modern Approach*. Pearson.

[12] Goodfellow, I., Bengio, Y., and Courville, A. *Deep Learning*. MIT Press.

## Glossary

$X_{r,t}$ Latent regional housing state.

$S$ Structural housing component.

$F$ Financial component.

$O$ Ownership component.

$L$ Legal component.

$Q$ Occupancy component.

$H$ Measurement matrix.

$A$ State transition matrix.

## The End